Exam :	End Term Quiz
Subject :	Maths2
Total Marks :	50.00
QP :	2024 Dec22: IIT M FN EXAM QDD2

Exam Mode Learning Mode

View Question Paper Summary

QUESTION MENU

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TIMER

00:26

CONTROLS

SUBMIT EXAM

Your Score

0.00 / 50.00

(0%)

Question 1 : 6406531045210 Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT **"FOUNDATION LEVEL : SEMESTER II: MATHEMATICS FOR DATA SCIENCE II (COMPUTER BASED EXAM)" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)**

OPTIONS :

- ◯ YES
- ◯ NO

Your score : 0

Discussions (0)

Question 2 : 6406531045211 View Solutions (1) Total Mark : 3.00 | Type : MSQ

Let $A \in M_{3\times3}(\mathbb{R})$ be a matrix with determinant 3. Which of the following operations transform A to a matrix B with determinant -6. Choose all the correct options:

OPTIONS :

- ☐ Multiply first row of A with 2 and interchange second and third row.
- ☐ Multiply matrix A with -2.
- ☐ Add third row to first row and multiply second row with -2.
- ☐ Interchange first and second row and multiply third row with -2.

Your score : 0

Discussions (0)

Question 3 : 6406531045215 View Solutions (0) Total Mark : 3.00 | Type : MSQ

Assume the usual vector addition and scalar multiplication for all the options below. Choose the correct option(s) from the following statements.

OPTIONS :

- [ ] The set of vectors $\{(x, y) \in \mathbb{R}^2 \mid x + y = 1\}$ forms a vector space.
- [ ] Let $A \in M_{3\times 3}(\mathbb{R})$ be a matrix such that the columns of A are linearly independent. The set of all solutions of the homogeneous system $Ax = 0$ is a vector space of dimension 0.
- [ ] The set of all $n \times n$ matrices with rank strictly less than n does not form a vector space.
- [ ] The set of all $n \times n$ matrices with determinant 0 forms a vector space.

Your score : 0

Discussions (0)

Question 4 : 6406531045236

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let f be a scalar-valued multivariable function defined on a domain $D \subset \mathbb{R}^2$. Among the following statements, choose all the options which guarantee that the directional derivative of f exists in all directions, at a given point $(a, b) \in D$.

OPTIONS :

- [ ] The gradient of f exists at (a, b).
- [ ] The function f is differentiable at (a, b).
- [ ] The gradient of f exists in a neighbourhood of (a, b) and is continuous at (a, b).
- [ ] The partial derivatives of f exist at (a, b).

Your score : 0

Discussions (0)

Question 5 : 6406531045247

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined as follows:

$$f(x, y) = \begin{cases} (x-y)^2 \sin\left(\frac{1}{x-y}\right), & \text{for } x \neq y \\ 0, & \text{for } x = y \end{cases}$$

Choose all the correct statements from the following.

OPTIONS :

- [ ] The function f is continuous at $(0, 0)$.
- [ ] There is a direction along which the directional derivative of f does not exist at $(0, 0)$.
- [ ] The equation of the tangent hyperplane to the graph of f at the point $(0, 0)$ is given by $z = 0$.
- [ ] The vector $(1, 1, -1)$ is orthogonal to the tangent hyperplane to the graph of f at $(0, 0)$.

Your score : 0

Discussions (0)

Question 6 : 6406531045212

Total Mark : 0.00 | Type : COMPREHENSION

Let $A = \begin{bmatrix} 1 & -1 & k \\ 1 & 1 & 1 \\ 5 & 1 & 3 \end{bmatrix}$ where $k \in \mathbb{R}$. Answer the given subquestions.

Your score : 0

Question 7 : 6406531045213

View Parent QN View Solutions (0)

Total Mark : 1.50 | Type : SA

Find the value of k for which the system $Ax = 0$ has infinitely many solutions.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)

Question 8 : 6406531045214

View Parent QN View Solutions (0)

Total Mark : 1.50 | Type : MCQ

If $k = 1$ and $b = \begin{bmatrix} 0 \\ 2 \\ \alpha \end{bmatrix}$, then which of the following option is true?

OPTIONS :

- If $\alpha = 5$, then the system $Ax = b$ will have a unique solution.

○ If $\alpha = 6$, then the system $Ax = b$ will have infinitely many solutions.

○ If $\alpha = 5$, then the rank of the augmented matrix $[A|b]$ is not equal to rank of A.

○ If $\alpha = 6$, then b does not belong to the column space of A.

Your score : 0

Discussions (0)

Question 9 : 6406531045216

Total Mark : 0.00 | Type : COMPREHENSION

Let $A = \begin{bmatrix} 3 & 1 & -2 \\ 1 & k & 1-k \end{bmatrix}$ where $k \in \mathbb{R}$. Answer the given subquestions as true or false:

Your score : 0

Question 10 : 6406531045217

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MCQ

The columns of A span $\mathbb{R}^2$ for all values of k.

OPTIONS :

○ TRUE

FALSE

Your score : 0

Discussions (0)

Question 11 : 6406531045218

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MCQ

The rank of A is 1 for some k.

OPTIONS :

TRUE

FALSE

Your score : 0

Discussions (0)

Question 12 : 6406531045219

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : MCQ

When $k = 1/3$, the row vectors span the plane in $\mathbb{R}^3$ given by the equation $x - 3y = 0$.

OPTIONS :

TRUE

FALSE

Your score : 0

Discussions (0)

Question 13 : 6406531045221

Total Mark : 0.00 | Type : COMPREHENSION

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear transformation and $A = \begin{bmatrix} 2 & 1 & 1 \\ 3 & 0 & 3 \\ -1 & -1 & 0 \end{bmatrix}$ is the matrix representation of T with respect to the standard ordered basis for both the domain and the codomain. Based on this information, answer the given subquestions:

Your score : 0

Question 14 : 6406531045222

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Choose the correct definition for T.

OPTIONS :

- $T(x, y, z) = (2x + 3y - z, x - z, x + 3y)$.
- $T(x, y, z) = (2x + y + z, 3x + 3z, -x - y)$.
- $T(x, y, z) = (x + y + z, 3x - 3z, -x - y - z)$.

Your score : 0

Discussions (0)

Question 15 : 6406531045223

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : SA

If B is the matrix representation of T with respect to the ordered basis $\{(1,2,0),(0,-1,3),(2,3,1)\}$ for both the domain and the codomain, find the rank of B.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)

Question 16 : 6406531045224

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the determinant of B.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)

Question 17 : 6406531045232

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data answer the given subquestions.

For a domain $D \subset \mathbb{R}^2$, let $f : D \to \mathbb{R}$ be the scalar valued function defined by the formula

$$f(x,y) = \frac{1}{x^2+y^2-1}, \text{ for all } (x,y) \in D.$$

Your score : 0

Question 18 : 6406531045233

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Which of the following is the largest possible set that would be a valid domain D for the given function f?

OPTIONS :

- $\mathbb{R}^2$
- $\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \neq 0\}$
- $\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \neq 1\}$
- $\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 < 1\}$

Your score : 0

Discussions (0)

Question 19 : 6406531045234

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MCQ

For $D = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 < 1\}$, what is the range of f?

OPTIONS :

- $\mathbb{R}$

- $\mathbb{R} \setminus \{0\}$
- $(0, \infty)$
- $(-\infty, -1]$

Your score : 0

Discussions (0)

Question 20 : 6406531045235

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MCQ

For $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 > 2\}$, what is the range of f?

OPTIONS :

- $(0, 1)$
- $\mathbb{R} \setminus \{0\}$
- $(0, \infty)$
- $(-\infty, -1]$

Your score : 0

Discussions (0)

Question 21 : 6406531045240

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data answer the given subquestions.

Consider the function $f(x, y) = e^{x^2y+xy+2y}$.

Your score : 0

Question 22 : 6406531045241

View Parent QN View Solutions (0)

Total Mark : 2.00 | Type : SA

If the gradient of f at the point $(3, 0)$ is given by (a, b), then what is the value of $b - a$?

Answer (Numeric):

Answer

Accepted Answer : 14

Your score : 0

Discussions (1)

Question 23 : 6406531045242

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the directional derivative of f at the point $(3, 0)$ in the direction $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$.

Answer (Numeric):

Answer

Accepted Answer : -7

Your score : 0

Discussions (0)

Question 24 : 6406531045248

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data answer the given subquestions.

Let x, y and z denote the length, breadth and height of a cuboid, respectively. Recall that the perimeter of the cuboid is given by $4(x+y+z)$ and the volume of the cuboid is given by xyz. Hence, perimeter and volume of an arbitrary cuboid are scalar valued functions in three variables (x, y and z) on the domain $\{(x, y, z) \in \mathbb{R}^3 \mid x > 0, y > 0, z > 0\}$.

Your score : 0

Question 25 : 6406531045249

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MCQ

Express the volume of a cuboid with perimeter 12, as a function ϕ of two variables x and y (length and breadth).

OPTIONS :

- $\phi(x, y) = xy(12 - x - y)$
- $\phi(x, y) = xy(12 + x + y)$
- $\phi(x, y) = xy(3 - x - y)$
- $\phi(x, y) = xy(3 + x + y)$

Your score : 0

Discussions (0)

Question 26 : 6406531045250

View Parent QN | View Solutions (0)

Total Mark : 2.00 | Type : SA

Using the function given, calculate the maximum possible volume of a cuboid whose perimeter is 12.

Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

Discussions (0)

Question 27 : 6406531045220

View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Consider the subspace $W = \{(x, y, z, w) \mid x + y = z, z + w = y - x\}$ of $\mathbb{R}^4$. Find the dimension of W.

OPTIONS :

- dim(W) = 1
- dim(W) = 2
- dim(W) = 3
- dim(W) = 4

Your score : 0

Discussions (0)

Question 28 : 6406531045225

Total Mark : 0.00 | Type : COMPREHENSION

Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation with range space of T given by $W = span\{(1, 2, -1), (3, 0, 1), (2, -2, 2)\}$. Based on the above information, answer the given subquestions:

Your score : 0

Question 29 : 6406531045226

View Parent QN | View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Let A be the matrix representation of T with respect to some ordered basis for both the domain and the codomain. Which of the following options is correct:

OPTIONS :

- The columns of A are always linearly independent.
- The rows of A may not be linearly dependent.
- The nullity of A is 0.
- The range of A forms a plane in $\mathbb{R}^3$.

Your score : 0

Discussions (0)

Question 30 : 6406531045227

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : MCQ

The transformation T is

OPTIONS :

- bijective
- injective but not surjective
- surjective but not injective
- neither injective nor surjective

Your score : 0

Discussions (0)

Question 31 : 6406531045228

Total Mark : 0.00 | Type : COMPREHENSION

Let $P_W : \mathbb{R}^3 \to \mathbb{R}^3$ be the projection (with respect to the standard inner product on $\mathbb{R}^3$) on the subspace $W = \{(x, y, z) \mid x - 2y + z = 0\}$. Based on the above information, answer the given subquestions:

Your score : 0

Question 32 : 6406531045229

View Parent QN View Solutions (0)

Total Mark : 1.50 | Type : SA

If $P_W(0, 1, 2) = (a, b, c)$, find $a + b + c$.

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

Discussions (0)

Question 33 : 6406531045230

View Parent QN | View Solutions (0)

Total Mark : 1.50 | Type : SA

If (α, β, γ) is a vector in the kernel of P_W, find $\alpha + \beta + \gamma$.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)

Question 34 : 6406531045231

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : SA

What is the dimension of the orthogonal complement $W^{\perp}$?

Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

Discussions (0)

Question 35 : 6406531045243

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data answer the given subquestions.

Consider the function $f(x, y) = x^3y + xy^2 + 2x + y^2$. Let $(1, a, b)$ be a point that lies on the tangent line to the graph of f at $(2, -1)$ in the direction of $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

Your score : 0

Question 36 : 6406531045244

View Parent QN | View Solutions (0)

Total Mark : 1.00 | Type : SA

If the directional derivative of f at the point $(2, -1)$ in the direction $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ is $\frac{k}{\sqrt{2}}$, then find the value of k.

Answer (Numeric):

Answer

Accepted Answer : 11

Your score : 0

Discussions (0)

Question 37 : 6406531045245

View Parent QN | View Solutions (0)

Total Mark : 1.50 | Type : SA

Find the value of a.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)

Question 38 : 6406531045246

View Parent QN View Solutions (0)

Total Mark : 1.50 | Type : SA

Find the value of b.

Answer (Numeric):

Answer

Accepted Answer : 10

Your score : 0

Discussions (0)

Question 39 : 6406531045251

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data answer the given subquestions.

Consider the function $f(x, y) = xy^2 - y^2 - x + 3$ defined on $\mathbb{R}^2$.

Your score : 0

Question 40 : 6406531045252

View Parent QN View Solutions (0)

Total Mark : 2.00 | Type : MCQ

How many critical points does the function f have?

OPTIONS :

- 0
- 1
- 2
- Infinitely many

Your score : 0

Discussions (0)

Question 41 : 6406531045253

View Parent QN View Solutions (0)

Total Mark : 2.00 | Type : MCQ

What can we conclude from the Hessian test for the function f?

OPTIONS :

- f does not have critical points.
- The Hessian test is inconclusive for all critical points.
- All of the critical points of f are local minima.
- All of the critical points of f are local maxima.
- All of the critical points of f are saddle points.

Your score : 0

Discussions (0)

Question 42 : 6406531045237

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data answer the given subquestions.

Let $f : \mathbb{R}^2 \setminus \{(0,0)\} \to \mathbb{R}$ be the function defined as follows:

$$f(x,y) = \frac{x^3y + xy^3 - x^2 - y^2 + k}{x^2 + y^2},$$

where $k \in \mathbb{R}$.

Your score : 0

Question 43 : 6406531045238

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the value of k for which the limit of f exists at $(0,0)$.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

Discussions (0)

Question 44 : 6406531045239

View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

For the given value of k, find the value of $f(0,0)$ for which f is a continuous function on $\mathbb{R}^2$.

Answer (Numeric):

Answer

Accepted Answer : -1

Your score : 0

Discussions (0)

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